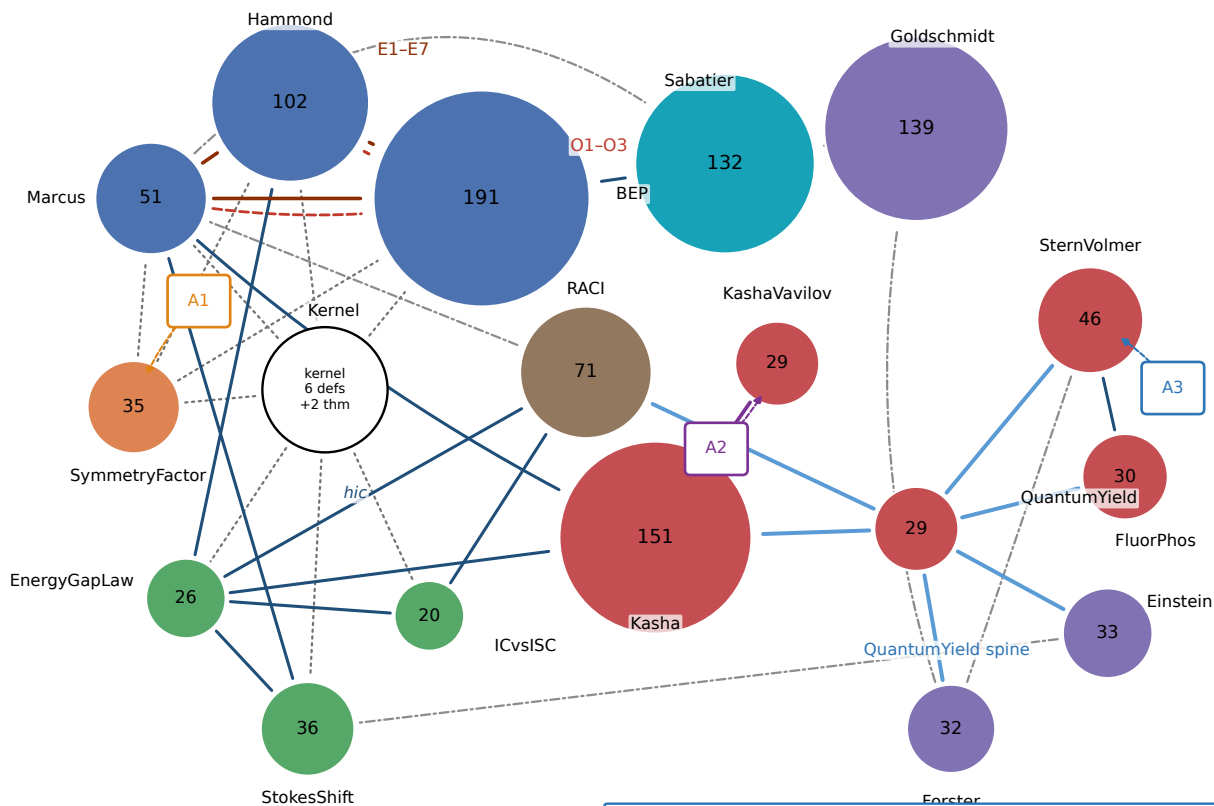




A3	static vs dynamic Stern-Volmer: the intensity channel is non-injective (matched curves coincide pointwise); the lifetime channel discriminates exactly: LifetimeTracks $m \leftrightarrow m = \text{Mech.dyn}$ ($0 < Ka$)
A2	Kasha vs Kasha-Vavilov: the four-conjunct d2_verdict — witnesses at $ic0 = 1 > 0$; coincidence iff under RateData + $0 < ic0$; the lossless corner $ic0 = 0$ separates them; boundary: kv_antikasha_boundary
A1	$\beta = 1/2$ vs the structural transfer coefficient: BetaHalfReading kr kp $\leftrightarrow$ kr = kp ($0 < kr, 0 < kp$) betaHalf_iff_equalForceConstants; witnesses $x^*(1, 4) = 2/3$, $x^*(4, 1) = 1/3$; unique in $[0, 1]$ without calculus